Checklists at the Speed of Prefill:

A Practical Recipe for Local, Calibrated Clinical Eligibility Screening with Open Decision Models

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September 2026

Abstract

Much of standard-of-care qualification means applying a published checklist to a free-text chart: screening eligibility, trial prescreening, protocol criteria. We show how to do this on one laptop with two open, prefill-only *decision models*, Laya (a 421M-parameter encoder) and Kev (a Qwen3.5 decoder with a LoRA adapter and a pointer head). These models return a calibrated probability for each typed yes/no question instead of generating text. The paper is organised as a seven-step recipe that a practitioner can follow for their own checklist. The steps are: write one question per criterion; choose a model; add deterministic arithmetic facts; combine criteria into an auditable conjunctive score; derive an automation threshold from a small labelled set; fine-tune locally if needed; and deploy with human review of the uncertain cases. We test each step on the USPSTF 2021 lung-cancer and 2019 AAA screening recommendations, using 900 held-out-template notes with exact per-criterion labels and 40 hand-written notes, and on 600 eligible-versus-excluded pairs from TREC Clinical Trials 2021. Zero-shot, Kev-4B is the strongest decision model (lung AUROC 0.88, AAA 1.00, TREC 0.82–0.84, on par with an 8B generative LLM on TREC), while Laya is near chance, as its authors warn. Two steps account for most of the gain. An 85-minute on-laptop RLCD fine-tune of the 421M Laya encoder on synthetic notes lifts lung AUROC on 40 hand-written notes from 0.55 to 0.90 (0.999 on held-out templates). The arithmetic hook then raises the share of hand-written notes that can be auto-decided at $\leq$5% error from 38% to 85% (AUROC 0.96), at 124 ms per note, about 10$\times$ faster than the generative baseline, which reaches AUROC 0.99 with the same hook. Thresholds measured on template notes do not transfer to hand-written ones, which is why Step 5 must use the practitioner's own notes. Code, a local screening app, data generators, per-item predictions and documentation are released under Apache-2.0.

1 Introduction

Clinical qualification work often reduces to a checklist over a chart. Examples are the USPSTF recommendation that adults aged 50–80 with a 20 pack-year smoking history be offered annual low-dose CT [Krist et al., 2021], or a trial's inclusion and exclusion criteria. The facts these checklists need often exist only in narrative text, such as "1 ppd since he was 18" or "quit after his MI in 2019". Rule engines over structured fields miss them. Generative large language models can read them [Jin et al., 2024, Wornow et al., 2025], but they are hard to run on premises. Their free-text output has to be parsed, and their confidence is not calibrated by default.

A newer class of open models answers *typed questions* about a text in a single prefill pass: *decision models*. Laya [Nandakishor M., 2026] and Kev [Palmer, 2026] both expose the `noul` (yes/no), `choice` and `score` interface of TypeSafe's System One API [TypeSafe, 2026]. They

return a probability per question and generate no text. For a checklist this interface is the right shape: one question per criterion, one probability each, and a principled way to decide which cases are safe to automate.

This paper is written to be *followed*. Section 3 is a seven-step recipe with code, which applies to any checklist. Section 4 tests each step on two USPSTF checklists and a public trial-matching benchmark, entirely on one Apple M1 Max laptop. Section 5 reports what worked, what did not, and how much each step contributes. Everything, including a point-and-click local screening app, is released (Section 7).

2 Decision models in one page

Laya [Nandakishor M., 2026] packs a question, its options (each introduced by a `[MASK]` marker) and the state into one sequence for a bidirectional encoder: ModernBERT-large [Warner et al., 2024], 421M parameters, English; or mmBERT-base, 322M, multilingual. A two-layer transformer head scores the hidden state at each marker. Training ("RLCD") combines a GRPO-style group policy gradient [Shao et al., 2024] on a reward built from strictly proper scoring rules [Gneiting and Raftery, 2007] with soft cross-entropy, and is followed by temperature scaling [Guo et al., 2017]. Its authors describe the released checkpoints as "a fast base to specialise, not a zero-shot decision engine".

Kev [Palmer, 2026] freezes a Qwen3.5 base (0.8B–9B) and trains a rank-16 LoRA [Hu et al., 2022] and a pointer head. The head scores each option's closing token against a final `<decide>` token. The note is prefilled once, and each question runs as an isolated row that reuses the cache. One temperature per checkpoint is fitted on held-out data. A Kev server speaks the System One HTTP API, so any client for that API works unchanged.

Why this matters clinically. Both models run on a workstation, so no protected health information leaves it. They answer from prefill only, so the cost of a decision does not grow with an explanation. And their probabilities can be thresholded, which is what makes selective automation possible [Geifman and El-Yaniv, 2017].

3 The recipe

We use lung-cancer screening as the running example. Each step names the file in the release that implements it.

Step 1 — Write one yes/no question per criterion

Split the recommendation into atomic criteria and write each as a `noul` question. Record the answer that eligibility *requires* (`clinical_system1/checklists.py`):

```
LUNG = Checklist(key="lcs", title="Lung cancer screening (USPSTF 2021)", criteria=(
  Criterion("lcs_age", "Is the patient between 50 and 80 years old, inclusive?"),
  Criterion("lcs_pack_years", "Does the patient have a smoking history of at least"
            " 20 pack-years (packs per day x years smoked; 20 cigarettes = 1 pack)?"),
  Criterion("lcs_recency", "Does the patient currently smoke, or did they quit"
            " 15 or fewer years ago?"),
  Criterion("lcs_limiting", "Does the patient have a health problem that substantially"
            " limits life expectancy, or are they unwilling or unable to undergo"
```

```
            " curative lung surgery?", required=False)))
```

Rules we found useful:

- state thresholds and inclusivity explicitly;
- define units in the question ("20 cigarettes = 1 pack");
- never combine two criteria in one question;
- phrase exclusions positively ("does X apply?") with `required=False`, rather than as a negated question.

Keep a *direct* question containing the whole recommendation as a baseline.

Step 2 — Choose a model

Start the model locally and send the questions:

```
cd upstream/kev && uv sync --extra serve
uv run --extra serve python -m kev.serve --run jaredpalmer/kev-4b --port 8009    # MLX on Apple,
    CUDA elsewhere
```

```
from clinical_system1.backends import KevBackend, LayaBackend, OllamaBackend
model = KevBackend(8009, "kev-4b")             # or LayaBackend(model="english", device="mps")
p = model.predict(note_text, LUNG.questions(include_direct=False))   # {criterion: P(yes)}
```

Our evidence (§5) gives a simple rule. For a zero-shot start, use **Kev-4B** on a laptop with $\geq$16 GB of unified memory or a GPU. For the best speed and accuracy, fine-tune **Laya** (Step 6); never use it zero-shot. A generative LLM, scored from its first-token Yes/No log-probabilities (`OllamaBackend`), is a strong but $\sim$10$\times$ slower reference.

Step 3 — Give the model the arithmetic

Decision models are weakest where the checklist needs arithmetic: packs/day $\times$ years, cigarettes to packs, a quit year against the visit year. A deterministic hook (`clinical_system1/facts.py`) finds these spans with generic regular expressions and appends the computed values; the model still makes every judgement:

```
Visit 2026. 58M ... Smokes ~1 ppd since he was 18 ...
[Computed facts: about 40 pack-years (1 packs/day x 40 years); age 58.]
```

If nothing matches, nothing is appended. This mirrors Kev's own `KEV_DATE_FACTS` option for day counts.

Step 4 — Combine criteria into an auditable score

Ask each criterion separately and combine them conjunctively:

$$P(\text{eligible}) = \prod_i \left[p_i \, \mathbf{1}(r_i = \text{yes}) + (1 - p_i) \, \mathbf{1}(r_i = \text{no}) \right]. \tag{1}$$

This treats the criteria as independent given the note, which is a heuristic. In return, every negative decision names the criterion that caused it (`Checklist.failed`), and a clinician can check that criterion alone. Always keep the per-criterion answers. Decide on your labelled set (Step 5) whether to threshold this product or the direct question. One miscalibrated criterion can drag the product down; for zero-shot Kev on hand-written notes, the direct question plus facts was the better score.

Step 5 — Label a small set and choose an automation threshold

Label 50–200 of your own notes for eligibility. Rank them by confidence $\max(P, 1-P)$ and find the lowest confidence τ at which the auto-decided notes stay within your error budget (`experiments/operating_points.py`). In production, notes with confidence $\geq \tau$ are auto-decided and the rest go to a clinician. The coverage at τ (the share of notes automated) is the number to plan staffing around. Never reuse a τ measured on one model, checklist, facts setting or note source for another.

Step 6 — If coverage is too low, fine-tune on your laptop

Write a seed generator for your checklist: sample the structured facts, render them as notes in several templates, and compute the labels exactly (`clinical_system1/vignettes.py`). Train on one template family and test on another. Our single-GPU port of Laya's RLCD recipe (`experiments/finetune_laya.py`) trains on 22,000 questions in about 85 minutes on an M1 Max and refits the temperature on a held-out slice:

```
upstream/laya/.venv/bin/python experiments/finetune_laya.py --epochs 2 --out runs/laya-clinical
```

Kev supports the same pattern (`kev.train --init_from jaredpalmer/kev-4b`), but its Mac training path is slow for Qwen3.5 bases. After fine-tuning, repeat Step 5: the threshold belongs to the model you ship.

Step 7 — Deploy with review

The release includes a local web app (`app/run.sh`, Fig. 3). It takes a note (or a batch), a checklist (built-in or edited in the browser), a model and an error budget, and returns per-criterion probabilities, P(eligible), the audit trail and an *auto-eligible / auto-ineligible / clinician review* decision. A "use in your code" tab exports the checklist as a System One request and as Python. For real use, add PHI redaction and audit logging (both upstreams provide prediction hooks). Have a clinician review a sample of automated negatives, and monitor coverage and error on a rolling labelled sample.

4 Evaluation

E1: USPSTF checklists. We encode lung-cancer screening [Krist et al., 2021] (four criteria) and AAA screening [Owens et al., 2019] (men 65–75 who have ever smoked; three criteria). Notes are rendered from structured seeds, so every label is exact. Smoking history is phrased three ways (packs/day × years, cigarettes/day, stated pack-years), and quitting two ways ("N years ago", "in YYYY"). Distractors include household smoking, secondhand exposure, family history and benign comorbidities. Life-limiting conditions include hospice-enrolled metastatic cancer, NYHA IV heart failure, oxygen-dependent COPD, advanced dementia, ESRD with dialysis declined, and explicit refusal of surgery. Threshold values are oversampled. Template family A (3,000 lung and 2,000 AAA notes) is used only for fine-tuning and calibration. Family B (500 lung and 400 AAA notes, 40% eligible) is used only for testing. We also wrote **40 free-text notes by hand**, with abbreviations, indirect dates and boundary cases (13 lung-eligible, 10 AAA-eligible).

E2: trial prescreening. From TREC Clinical Trials 2021 [Roberts et al., 2021] we take 600 judged patient–trial pairs, eligible versus excluded (300 each, stratified over the 75 topics), with eligibility text from ClinicalTrials.gov. We ask a direct question and "meets the inclusion criteria?" / "meets any exclusion criterion?", and score $P(\text{inc})(1 - P(\text{exc}))$.

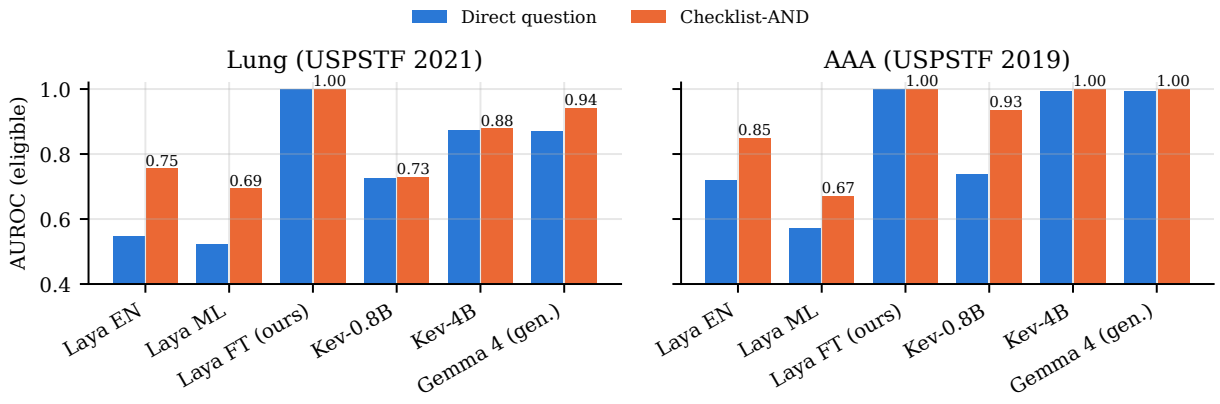


Figure 1: Eligibility AUROC on held-out-template notes: one direct question (blue) versus the checklist-AND product (orange).

Table 1: USPSTF 2021 lung cancer screening eligibility on 500 held-out-template notes (40% eligible). Coverage = share of notes auto-decided at the given error budget. Bold = best.

	Direct	Checklist-AND					All crit.	
Model	AUROC	AUROC	Acc	Brier↓	Cov@5%	Cov@10%	correct	n
Laya EN (0-shot)	0.547	0.754	0.600	0.369	0.130	0.168	0.082	500
Laya ML (0-shot)	0.523	0.693	0.608	0.284	0.000	0.130	0.242	500
Laya EN + RLCD FT (ours)	**0.998**	**0.999**	**0.992**	**0.008**	**1.000**	**1.000**	**0.972**	500
Kev-0.8B	0.726	0.728	0.602	0.260	0.268	0.316	0.288	500
Kev-4B	0.872	0.877	0.648	0.244	0.424	0.492	0.402	500
Gemma 4 (generative)	0.870	0.940	0.888	0.085	0.776	0.924	0.712	250

Systems. All run locally on an Apple M1 Max (64 GB):

- Laya EN and Laya ML (released checkpoints, MPS);
- Kev-0.8B and Kev-4B (released adapters, MLX bf16);
- Laya EN after our local RLCD fine-tune;
- Gemma 4 through Ollama (∼8B, 4-bit), asked one question at a time, with P(yes) from the first-token Yes/No log-probabilities. For cost reasons it is run on the first 250 notes per E1 split and on a balanced 300-pair E2 subset.

Metrics. We report AUROC (with 1,000-resample bootstrap CIs), accuracy, Brier score, ECE, coverage at 5% and 10% error budgets, the share of notes with every criterion right, and uncontended latency per note for the full checklist.

5 Results

Step 2: which model zero-shot? Tables 1–6 answer this. **Kev-4B** is the strongest decision model out of the box: lung AUROC 0.877 (checklist-AND), AAA 1.000, and TREC 0.82–0.84. On TREC it matches the 8B generative baseline (0.80) at roughly half the latency. Kev-0.8B is clearly weaker (lung 0.73, TREC 0.63): the Qwen base carries most of the capability. **Laya zero-shot is not usable** for this task. Lung AUROC is 0.55–0.75, and on only 8% of notes are all four criteria

Table 2: Lung screening on 40 hand-written free-text notes (13 eligible).

	Direct	Checklist-AND					All crit.	
Model	AUROC	AUROC	Acc	Brier↓	Cov@5%	Cov@10%	correct	n
Laya EN (0-shot)	0.462	0.553	0.675	0.302	0.075	0.075	0.100	40
Laya ML (0-shot)	0.590	0.293	0.675	0.302	0.025	0.025	0.175	40
Laya EN + RLCD FT (ours)	**0.913**	**0.903**	**0.875**	**0.122**	**0.375**	**0.750**	**0.850**	40
Kev-0.8B	0.821	0.553	0.675	0.238	0.100	0.100	0.375	40
Kev-4B	0.869	0.732	0.650	0.254	0.275	0.325	0.500	40
Gemma 4 (generative)	0.838	0.840	0.800	0.163	0.250	0.600	0.650	40

Table 3: Per-criterion accuracy on the lung-screening test set (500 notes, template family B).

Model	Age 50–80	≥20 pack-yr	Recency	Limiting
Laya EN (0-shot)	0.444	0.592	0.486	0.562
Laya ML (0-shot)	0.708	0.634	0.520	0.878
Laya EN + RLCD FT (ours)	1.000	0.992	0.980	1.000
Kev-0.8B	0.722	0.686	0.622	0.922
Kev-4B	0.982	0.774	0.682	0.830
Gemma 4 (generative)	0.968	0.840	0.888	0.976

right; it even misjudges the plain age criterion (44% accuracy). This matches upstream's warning that the `noul` head can follow its label text instead of the state. Laya-multilingual is at chance on TREC even with an 8,192-token window. The arithmetic criteria are the bottleneck for every zero-shot model (Table 3). For Kev-4B, age is 98% correct but pack-years only 77% and recency 68%. The explicit-age AAA checklist is essentially solved (Table 5).

Step 3: arithmetic facts. The deterministic hook (Table 4) raises pack-year accuracy for every model. Kev-4B goes from 0.77 to 0.98 on template notes and from 0.82 to 1.00 on hand-written notes, and Gemma 4 goes from 0.84 to 0.99. It helps the fine-tuned Laya most where it matters, on hand-written notes: AUROC rises from 0.903 to 0.957, and coverage at a 5% error budget from 38% to 85%. One gap remains. Kev's *recency* accuracy does not move (0.68) even when "quit about 16 years before the visit" is stated, so Kev does not reliably compare a stated number with the 15-year cut-off inside a disjunctive question. The hook computed pack-years exactly for all 433 template smokers, and for 34 of the 36 hand-written smokers. It missed "from her 20s until 2005" and "a few cigarettes on weekends". It also did not turn "from age 16 to age 60" into a quit interval.

Step 4: direct question or checklist-AND? On template notes the conjunctive score is at least as good as the direct question for every system: Gemma 4 lung 0.870→0.940, and Kev-0.8B AAA 0.736→0.934. It also gives an audit trail that is fully correct on 97% of notes for the fine-tuned Laya. It is not uniformly better, however. For zero-shot Kev on hand-written lung notes the direct question wins (Kev-4B 0.869 vs 0.732; with facts 0.966 vs 0.764). The product is dragged down by one miscalibrated criterion (recency), giving a sensitivity of only 0.15. Our advice is therefore to keep the per-criterion answers for audit, but to choose the score you threshold on your own labelled set.

Table 4: Step 3 ablation: the computed-facts hook on the lung checklist (checklist-AND). Δ = with minus without.

Notes	Model	AUROC	Pack-yr acc	Recency acc	Cov@5%	All crit.
Template	Kev-0.8B	0.81 (+0.08)	0.84 (+0.16)	0.63 (+0.01)	0.35 (+0.08)	0.34 (+0.06)
Template	Kev-4B	0.94 (+0.06)	0.98 (+0.20)	0.68 (+0.00)	0.52 (+0.10)	0.44 (+0.04)
Template	Laya EN (0-shot)	0.70 (-0.06)	0.75 (+0.16)	0.70 (+0.21)	0.18 (+0.05)	0.11 (+0.03)
Template	Laya EN + RLCD FT (ours)	0.99 (-0.00)	1.00 (+0.01)	0.94 (-0.04)	1.00 (+0.00)	0.93 (-0.04)
Template	Gemma 4 (generative)	1.00 (+0.06)	0.99 (+0.15)	0.92 (+0.03)	1.00 (+0.22)	0.88 (+0.16)
Hand-written	Kev-0.8B	0.64 (+0.09)	0.85 (+0.05)	0.70 (-0.03)	0.15 (+0.05)	0.42 (+0.05)
Hand-written	Kev-4B	0.76 (+0.03)	1.00 (+0.18)	0.68 (+0.00)	0.25 (-0.03)	0.55 (+0.05)
Hand-written	Laya EN (0-shot)	0.76 (+0.21)	0.78 (+0.22)	0.62 (+0.10)	0.40 (+0.33)	0.20 (+0.10)
Hand-written	Laya EN + RLCD FT (ours)	0.96 (+0.05)	0.95 (+0.10)	0.93 (-0.05)	0.85 (+0.47)	0.88 (+0.03)
Hand-written	Gemma 4 (generative)	0.99 (+0.15)	0.97 (+0.12)	0.82 (+0.05)	0.80 (+0.55)	0.78 (+0.12)

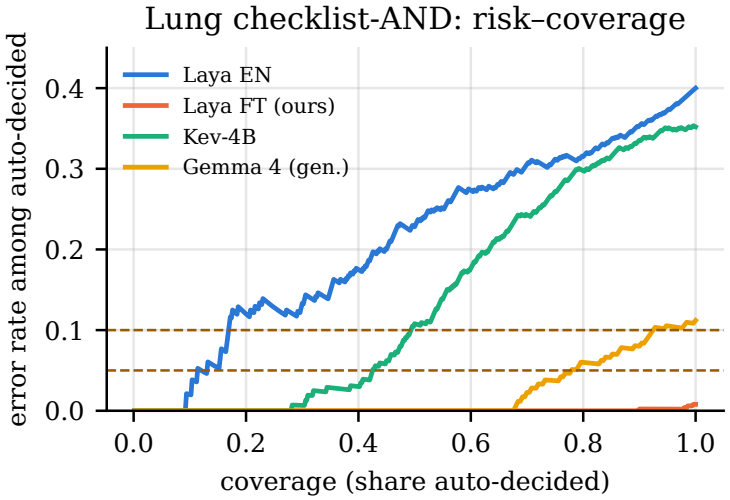


Figure 2: Risk–coverage for the lung checklist-AND score on held-out-template notes. A curve that stays under the dashed error budget longer automates more notes.

Step 5: thresholds do not transfer. Coverage at a 5% error budget is the most practical number here, and also the most fragile. The fine-tuned Laya automates 100% of template notes at ≤5% error with τ=0.98, but only 38% of hand-written notes (85% with the facts hook). Kev-4B automates 42% of template notes but 28% of hand-written ones. A threshold measured on synthetic notes would badly overstate what is safe on real text. This is why the recipe measures τ on the practitioner's own labelled notes.

Step 6: local fine-tuning. Our single-MPS port of Laya's RLCD recipe trained on 22,000 family-A questions in 85 minutes. It fitted a `noul` temperature of 3.5 on a held-out slice (0.996 accuracy there). On the unseen family-B templates it reaches lung AUROC 0.999, AAA 1.000, Brier 0.008, with every criterion correct on 97% of notes. Because the two template families share phrase vocabulary, the hand-written notes are the honest test. There it scores lung AUROC 0.903 (0.957 with facts), with every criterion correct on 85% (87.5%) of notes, against 0.553 and 10% before fine-tuning. A small encoder, specialised on a laptop with synthetic data, becomes the best *decision* model in the study.

Table 5: USPSTF 2019 AAA screening eligibility on 400 held-out-template notes (40% eligible).

	Direct	Checklist-AND					All crit.	
Model	AUROC	AUROC	Acc	Brier↓	Cov@5%	Cov@10%	correct	n
Laya EN (0-shot)	0.720	0.849	0.600	0.300	0.417	0.468	0.285	400
Laya ML (0-shot)	0.571	0.669	0.630	0.229	0.007	0.028	0.355	400
Laya EN + RLCD FT (ours)	**1.000**	**1.000**	**1.000**	0.000	**1.000**	**1.000**	**1.000**	400
Kev-0.8B	0.736	0.934	0.848	0.131	0.527	0.777	0.613	400
Kev-4B	0.993	1.000	0.988	0.017	**1.000**	**1.000**	0.965	400
Gemma 4 (generative)	0.994	**1.000**	**1.000**	**0.000**	**1.000**	**1.000**	0.996	250

Table 6: TREC Clinical Trials 2021, eligible vs. excluded (balanced; 95% bootstrap CI for AUROC).

	Direct		Inc.×(1−Exc.)		
Model	AUROC	Bal. acc	AUROC	Bal. acc	n
Laya ML (8k ctx)	0.514 [0.47, 0.56]	0.503	0.481 [0.43, 0.53]	0.500	600
Kev-0.8B	0.633 [0.59, 0.68]	0.515	0.627 [0.58, 0.67]	0.587	600
Kev-4B	0.820 [0.78, 0.85]	0.732	0.839 [0.81, 0.87]	0.652	600
Gemma 4 (generative)	0.803 [0.75, 0.85]	0.560	0.797 [0.74, 0.84]	0.623	300

E2: trial prescreening. Distinguishing eligible from excluded patients for a trial they are topically relevant to is harder (Table 6). Kev-4B reaches AUROC 0.820 [0.78, 0.85] directly and 0.839 [0.81, 0.87] with the inclusion×(1−exclusion) decomposition. Gemma 4 reaches 0.803 and 0.797. Coverage at 5% error is low for all systems (≤7%), so trial prescreening is a triage aid here, not something to automate.

Latency. Uncontended, on one M1 Max, for the full five-question lung checklist (Table 7): the fine-tuned Laya takes 124 ms per note on the GPU and 506 ms on CPU alone. Kev-0.8B takes 134 ms, Kev-4B 694 ms and Gemma 4 1,250 ms. Ollama's prompt caching narrows the gap, but the generative model still needs one decode per question. At 124 ms per note, a single laptop screens about 29,000 notes per hour.

6 Discussion

What a practitioner should take away. (i) The typed-question interface maps onto clinical checklists with no glue code, and it produces an audit trail per criterion. (ii) Out of the box, a 4B decision model is a reasonable prescreener for explicit criteria (age, sex, ever-smoked) and for trial triage. It is not reliable for criteria that need arithmetic. (iii) Two cheap, local steps close most of the gap. The first is a deterministic hook that states the arithmetic; the second is an 85-minute fine-tune of a 421M encoder on synthetic notes from a seed generator. Together they give the fastest system in the study and, on hand-written notes, the second most accurate after a 10× slower generative model given the same hook. (iv) The automation threshold must be measured on the target notes. A threshold taken from synthetic data overstated safe coverage by a factor of 2.6 in our study (100% vs 38%).

Where generative models still win. With the facts hook, the 8B generative baseline was the most accurate system on hand-written lung notes (AUROC 0.986). It reads varied phrasing better

Table 7: Latency for the full 5-question lung checklist per note on one Apple M1 Max, uncontended (50 notes after 3 warm-up calls).

System	Params	p50 ms	p90 ms	notes/s
Laya EN, Apple GPU (MPS)	421M	126	136	7.96
Laya EN, CPU only	421M	506	571	1.93
Laya FT (ours), MPS	421M	124	136	8.04
Kev-0.8B, MLX bf16	0.8B	134	137	7.68
Kev-4B, MLX bf16	4B	694	703	1.49
Gemma 4, Ollama	∼8B	1250	1373	0.79

than any decision model we tested. If latency and throughput matter less than accuracy, and the model can run on premises, first-token log-probabilities from a generative model are a strong alternative. They slot into the same recipe, and the app supports them.

Limitations. The E1 notes are template-generated. Disjoint template families and the hand-written notes narrow the gap to real EHR text but do not close it. The hand-written set is small, and it was written and labelled by the same author who wrote the facts hook, which may flatter the hook. TREC topics are synthetic, and TREC relevance labels are noisy. The conjunctive score assumes independence. We ran small decision models only ($\leq$4B), and a single 4-bit generative baseline without chain-of-thought. Nothing here is validated for patient care. Deployment needs prospective evaluation on real charts, subgroup audits, and clinician review of automated negatives.

7 Release

https://github.com/menonpg/clinical-system1 (Apache-2.0) contains:

- the `clinical_system1` package (checklists, vignette generator, facts hook, backends, metrics);
- every experiment script and per-item prediction;
- the local fine-tuning port;
- the screening app;
- HTML documentation of Laya and Kev and of this study.

Upstreams: https://github.com/NandhaKishorM/laya, https://github.com/jaredpalmer/kev.

References

Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, 2017.

Edward J. Hu, Yelong Shen, Phillip Wallis, et al. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

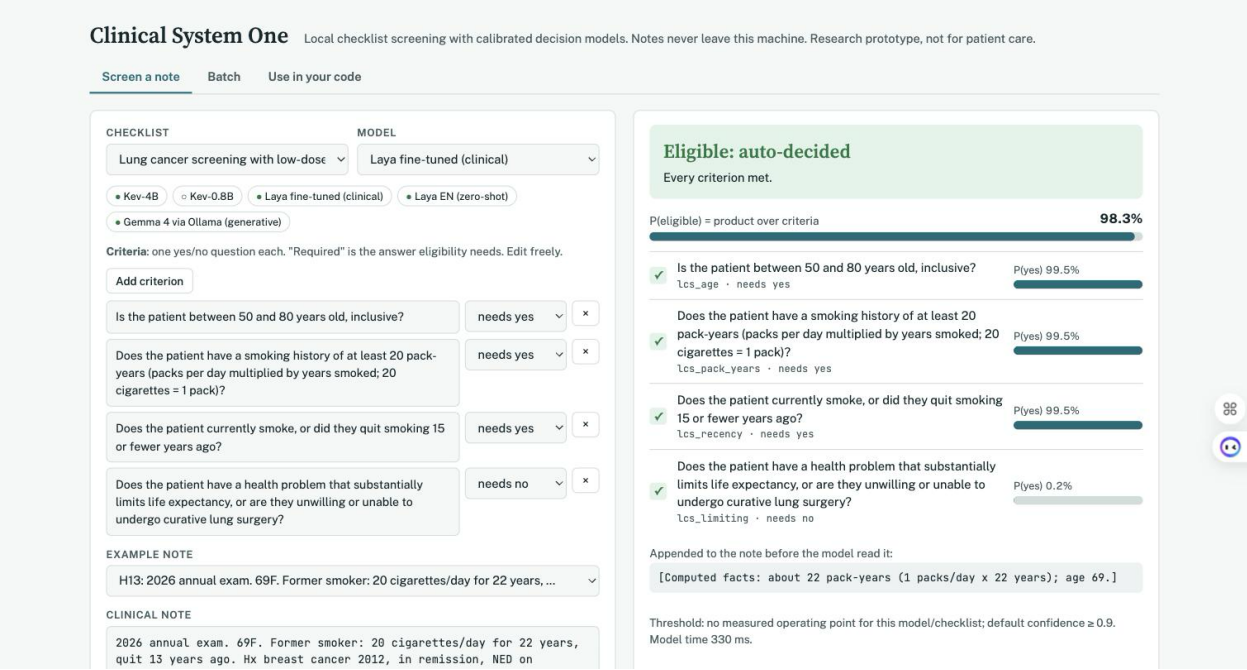


Figure 3: The released local screening app. The note, checklist and model are on the left. On the right are per-criterion $P(\text{yes})$, the conjunctive $P(\text{eligible})$, the computed facts that were appended, and the automation decision at the chosen error budget.

Qiao Jin, Zifeng Wang, Charalampos S. Floudas, et al. Matching patients to clinical trials with large language models. *Nature Communications*, 15:9074, 2024.

Alex H. Krist, Karina W. Davidson, Carol M. Mangione, et al. Screening for lung cancer: US preventive services task force recommendation statement. *JAMA*, 325(10):962–970, 2021.

Nandakishor M. Laya: Multilingual, non-autoregressive system 1 decision engine. https://github.com/NandhaKishorM/laya, 2026. Version 0.3.20, Apache-2.0.

Douglas K. Owens, Karina W. Davidson, Alex H. Krist, et al. Screening for abdominal aortic aneurysm: US preventive services task force recommendation statement. *JAMA*, 322(22):2211–2218, 2019.

Jared Palmer. Kev: Small decision models you can train and run yourself. https://github.com/jaredpalmer/kev, 2026. Apache-2.0.

Kirk Roberts, Dina Demner-Fushman, Ellen M. Voorhees, Steven Bedrick, and William R. Hersh. Overview of the TREC 2021 clinical trials track. In *Proceedings of the Thirtieth Text REtrieval Conference (TREC 2021)*, 2021.

Zhihong Shao, Peiyi Wang, Qihao Zhu, et al. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

TypeSafe. System one API reference. https://docs.typesafe.ai/api, 2026.

Benjamin Warner, Antoine Chaffin, Benjamin Clavié, et al. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv preprint arXiv:2412.13663*, 2024.

Michael Wornow, Alejandro Lozano, Dev Dash, Jenelle Jindal, Kenneth W. Mahaffey, and Nigam H. Shah. Zero-shot clinical trial patient matching with LLMs. *NEJM AI*, 2(1), 2025.

A Question wording

Lung cancer screening with low-dose CT (USPSTF 2021 (Grade B), JAMA 2021;325(10):962-970).

- `lcs_age` (required: yes): Is the patient between 50 and 80 years old, inclusive?
- `lcs_pack_years` (required: yes): Does the patient have a smoking history of at least 20 pack-years (packs per day multiplied by years smoked; 20 cigarettes = 1 pack)?
- `lcs_recency` (required: yes): Does the patient currently smoke, or did they quit smoking 15 or fewer years ago?
- `lcs_limiting` (required: no): Does the patient have a health problem that substantially limits life expectancy, or are they unwilling or unable to undergo curative lung surgery?
- `lcs_eligible` (direct): Under the USPSTF 2021 lung cancer screening recommendation (adults aged 50 to 80 years who have a 20 pack-year smoking history and currently smoke or have quit within the past 15 years, and who do not have a health problem that substantially limits life expectancy or the ability or willingness to have curative lung surgery), is this patient eligible for annual low-dose CT screening?

One-time abdominal aortic aneurysm ultrasound screening (USPSTF 2019 (Grade B), JAMA 2019;322(22):2211-2218).

- `aaa_male` (required: yes): Is the patient male?
- `aaa_age` (required: yes): Is the patient between 65 and 75 years old, inclusive?
- `aaa_ever_smoked` (required: yes): Has the patient ever smoked tobacco (current or former smoker)?
- `aaa_eligible` (direct): Under the USPSTF 2019 recommendation (one-time ultrasound screening for abdominal aortic aneurysm in men aged 65 to 75 years who have ever smoked), is this patient eligible for AAA screening?

TREC CT 2021.

- `eligible`: Based on the patient description, is this patient eligible for the clinical trial, meaning they satisfy the inclusion criteria and meet none of the exclusion criteria?
- `meets_inclusion`: Does the patient satisfy the trial's inclusion criteria?
- `meets_exclusion`: Does the patient meet any of the trial's exclusion criteria?

B Reproducing every number

```
experiments/prep_trec.py            # TREC CT 2021 sample + ClinicalTrials.gov criteria
experiments/run_all.sh              # all models, one at a time, resumable
experiments/run_facts.sh            # Step 3 ablation
experiments/analyze.py              # summary.json
experiments/figures.py; paper/make_tables.py
```